

Elisaweta Masserova

elisawem@cs.cmu.edu

EDUCATION

2018 – 2024	Ph.D. student: Computer Science Department Advised by Prof. Bryan Parno and Prof. Vipul Goyal Carnegie Mellon University
2016 – 2019	Master studies: Informatics Karlsruhe Institute of Technology
2013 – 2018	Bachelor studies: Mathematics Karlsruhe Institute of Technology
2011 – 2016	Bachelor studies: Informatics Karlsruhe Institute of Technology

WORK AND RESEARCH EXPERIENCE

2024-ongoing	Computer Science Department, Carnegie Mellon University Post Doctoral Fellow • <i>Research under the guidance of Prof. Elaine Shi</i>
2024	JP Morgan Chase Summer Research Associate Intern • <i>Designed a scheme for proving fairness of decision trees</i>
2023	JP Morgan Chase Summer Research Associate Intern • <i>Proved theorem for communication complexity of stateless multi-party computation</i>
2019, 2020	Computer Science Department, Carnegie Mellon University Teaching assistant (part of Ph.D. program requirements) • <i>Grading student assignments</i> • <i>Suggested homework problem statements</i> • <i>Improving lecture materials</i>
2014 – 2018	Freelancing in software- and web-development • <i>Developed multiple websites</i>
2014 – 2018	Institute of Theoretical Informatics at KIT, student research assistant • <i>Developed a visualization of a zero-knowledge proof example for educational purposes</i> • <i>Designed various educational materials, posters, flyers</i> • <i>Worked on the topic of CCA-secure commitments</i>
2013 – 2014	bluehands GmbH & Co.munication KG, software development • <i>Collaborated on the development of a city guide app</i>
2013	FZI Research Center for Information Technology, applied software development • <i>Developed a training app to support the recovery of stroke patients</i>

HONORS AND AWARDS

2024	Carnegie Bosch Institute (CBI) Postdoctoral Fellowship
2022	Protocol Labs grant for the “Stateless Distributed Randomness Generation” project
2015	InterAct scholarship for bachelor thesis research at CMU, Pittsburgh USA
2014	1st Place at the HackZurich, Europe’s largest hackathon (Team „Immersive“) • <i>Virtual reality app to capture and view immersive panorama photos</i>
2014	1st Place at the Microsoft Imagine Cup national Final (Team „Krowd“) • <i>System to help improve crowd security at big events, e.g., concerts, festivals, etc.</i>
2014 – 2015	Participated in the Femtec Careerbuilding-Program • <i>Stipend for outstanding females in engineering and applied sciences, in cooperation with Germany’s top Universities and Employers</i>
2012 – 2014	ACM ICPC German Collegiate Programming Contest (GCPC) participation ACM ICPC Northwestern European Regional Contest (NWERC) participation
2013	App Art Award by ZKM Karlsruhe (Team “Bitphonie”) • <i>App that converted live sensor data of smartphones into music</i>
2005 – 2010	Multiple awards at the Saporoshje city Olympiads in mathematics and physics
2009	8th place at the Ukrainian finals of the International competition for mathematical and logical games

PUBLICATIONS

2024	Non-Interactive and Publicly Verifiable Zero-Knowledge Proof for Fair Decision Trees <i>Elisaweta Masserova, Antigoni Polychroniadou, Akira Takahashi</i> • <i>Regml Workshop at Annual Conference on Neural Information Processing Systems 2024</i> • Preprint will be available soon
2024	Randomness from Minimal Assumptions where Parties Speak Sequentially Once <i>Chen-Da Liu-Zhang, Elisaweta Masserova, João Ribeiro, Pratik Soni, Sri AravindaKrishnan Thyagarajan</i> • <i>In submission</i> • Preprint will be available soon
2024	Improved YOSO Randomness Generation with Worst-Case Corruptions <i>Chen-Da Liu-Zhang, Elisaweta Masserova, João Ribeiro, Pratik Soni, Sri AravindaKrishnan Thyagarajan</i> • <i>Financial Cryptography and Data Security (FC) 2024</i> • https://eprint.iacr.org/2024/1268
2023	Towards YOSO MPC via Packed Secret-Sharing <i>Elisaweta Masserova, Antigoni Polychroniadou, Daniel Escudero</i> • <i>In submission, preprint will be available soon</i>
2022	Rapidash: Foundations of Side-Contract-Resilient Fair Exchange <i>Hao Chung, Elisaweta Masserova, Elaine Shi, Sri AravindaKrishnan Thyagarajan</i> • <i>Science of Blockchain Conference (CBS) 2024</i> • https://eprint.iacr.org/2022/1063

2022	Ponyta: Foundations of Side-Contract-Resilient Fair Exchange <i>Hao Chung, Elisaweta Masserova, Elaine Shi, Sri AravindaKrishnan Thyagarajan</i> • <i>Crypto Economics Security Conference (CESC) 2022</i> • https://eprint.iacr.org/2022/545
2022	Logic Locking – Connecting Theory and Practice <i>Elisaweta Masserova, Deepali Garg, Ken Mai, Lawrence Pileggi, Vipul Goyal, Bryan Parno</i> • <i>In submission</i> • https://eprint.iacr.org/2022/545
2021	Blockchains Enable Non-Interactive MPC <i>Vipul Goyal, Elisaweta Masserova, Bryan Parno, Yifan Song</i> • <i>Theory of Cryptography Conference (TCC) 2021</i> • https://eprint.iacr.org/2021/1233
2020	Storing and Retrieving Secrets on a Blockchain (Fast Batched DPSS and its Applications) <i>Vipul Goyal, Abhiram Kothapalli, Elisaweta Masserova, Bryan Parno, Yifan Song</i> • <i>International Conference on Practice and Theory in Public Key Cryptography (PKC) 2022</i> • https://eprint.iacr.org/2020/504
2020	Poppins: A Direct Construction for Asymptotically Optimal zkSNARKs <i>Abhiram Kothapalli, Elisaweta Masserova, Bryan Parno</i> • ePrint Archive, Report 2020/1318 • https://eprint.iacr.org/2020/1318
2020	Talek: Private Group Messaging with Hidden Access Patterns <i>Raymond Cheng, William Scott, Elisaweta Masserova, Irene Zhang, Vipul Goyal, Thomas Anderson, Arvind Krishnamurthy, Bryan Parno</i> • <i>Annual Computer Security Applications Conference (ACSAC) 2020</i> • https://eprint.iacr.org/2020/066
2014	sGrid: Centralized management of BOINC-based research projects – Summary <i>Emanuel Jöbstl, Jerome Urhausen, Elisaweta Masserova, Ainara Askar</i> <i>Poster, In GI-Jahrestagung, p. 2487. (2014)</i>

THESIS

2024	Distributed Cryptography as a Service <i>Ph.D. thesis, Advisors: Prof. Bryan Parno, Prof. Vipul Goyal</i>
2018	Rerandomizing Oblivious Transfer for Online Oblivious Bingo Voting <i>Master thesis, Advisors: Prof. Müller-Quade, Rebecca Schwerdt, M.Sc.</i>
2016	Evaluation of Publishable Humanly Usable Secure Password Creation Schemas <i>Bachelor thesis, Advisors: Prof. Manuel Blum and Prof. Müller-Quade</i>

SERVICE

Program committee member	FC 2024, CCS 2024, Usenix 2025, The Web Conference 2025, FC 2025
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ADDITIONAL INFORMATION

Extracurricular activities	Member of Studies Without Borders student organisation, member of IACR
Language skills	Mother tongue Ukrainian and Russian Fluent German and English
Spare time interests	Digital art, photography

Pittsburgh, 10.10.2024